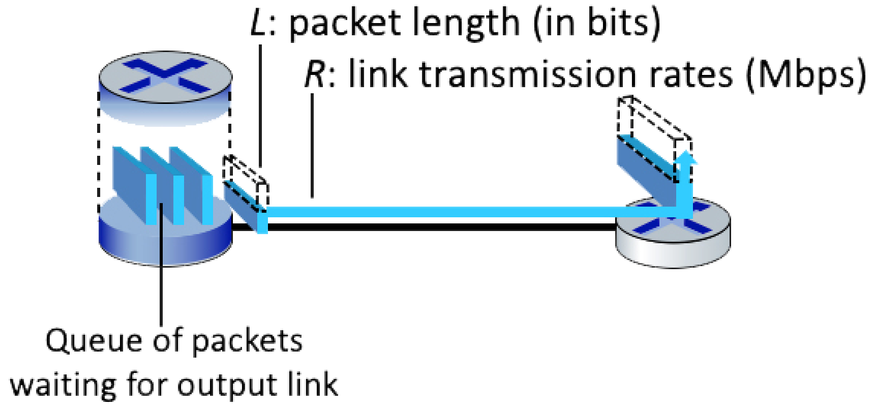
INTERACTIVE END-OF-CHAPTER EXERCISES

COMPUTING THE ONE-HOP TRANSMISSION DELAY

Consider the figure below, in which a single router is transmitting packets, each of length *L* bits, over a single link with transmission rate *R* Mbps to another router at the other end of the link.



Suppose that the packet length is *L*= 4000 bits, and that the link transmission rate along the link to router on the right is *R* = 10 Mbps.

Round your answer to two decimals after leading zeros

QUESTION LIST

1. What is the transmission delay?

2. What is the maximum number of packets per second that can be transmitted by this link?

SOLUTION

The transmission delay = L/R = 4000 bits / 10000000 bps = 0.0004 seconds

The number of packets that can be transmitted in a second into the link = R / L = 10000000 bps / 4000 bits = 2500 packets

QUESTION 1 OF 2

What is the transmission delay?

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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